

Problem 13.6.1

Show that

a) $\sinh z = \sinh x \cos y + i \cosh x \sin y$

b) $\cosh z = \cosh x \cos y + i \sinh x \sin y$

Solution

a)

$$\begin{aligned}
 \sinh z &= \frac{1}{2}(e^z - e^{-z}) \\
 &= \frac{1}{2}(e^{x+iy} - e^{-x-iy}) \\
 &= \frac{1}{2}(e^x e^{iy} - e^{-x} e^{-iy}) \\
 &= \frac{1}{2}(e^x (\cos y + i \sin y) - e^{-x} (\cos y - i \sin y)) \\
 &= \frac{1}{2}(e^x \cos y + i e^x \sin y - e^{-x} \cos y + i e^{-x} \sin y) \\
 &= \frac{1}{2}((e^x - e^{-x}) \cos y + i(e^x + e^{-x}) \sin y) \\
 &= \boxed{\sinh x \cos y + i \cosh x \sin y}
 \end{aligned}$$

b)

$$\begin{aligned}
 \cosh z &= \frac{1}{2}(e^z + e^{-z}) \\
 &= \frac{1}{2}(e^{x+iy} + e^{-x-iy}) \\
 &= \frac{1}{2}(e^x e^{iy} + e^{-x} e^{-iy}) \\
 &= \frac{1}{2}(e^x (\cos y + i \sin y) + e^{-x} (\cos y - i \sin y)) \\
 &= \frac{1}{2}(e^x \cos y + i e^x \sin y + e^{-x} \cos y - i e^{-x} \sin y) \\
 &= \frac{1}{2}((e^x + e^{-x}) \cos y + i(e^x - e^{-x}) \sin y) \\
 &= \boxed{\cosh x \cos y + i \sinh x \sin y}
 \end{aligned}$$

Problem 13.6.3

Show that

a) $\cosh^2 z - \sinh^2 z = 1$

b) $\cosh^2 z + \sinh^2 z = \cosh 2z$

Solution

a)

$$\begin{aligned}\cosh^2 z - \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 - \frac{1}{4}(e^z - e^{-z})^2 \\ &= \frac{1}{4}[(e^{2z} + 2 + e^{-2z}) - (e^{2z} - 2 + e^{-2z})] \\ &= \boxed{1}\end{aligned}$$

b)

$$\begin{aligned}\cosh^2 z + \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 + \frac{1}{4}(e^z - e^{-z})^2 \\ &= \frac{1}{4}[(e^{2z} + 2 + e^{-2z}) + (e^{2z} - 2 + e^{-2z})] \\ &= \frac{1}{4}(2e^{2z} + 2e^{-2z}) \\ &= \frac{1}{2}(e^{2z} + e^{-2z}) \\ &= \boxed{\cosh 2z}\end{aligned}$$

Problem 13.6.7Find, in the form $u + iv$

a) $\cos i$

b) $\sin i$

Solution**Problem 13.6.13**

Using the definitions, prove

a) $\cos z$ is even, $\cos(-z) = \cos z$

b) $\sin z$ is odd, $\sin(-z) = -\sin z$

Solution**Problem 13.6.16**

Find all solutions

$$\sin z = 100$$

Solution**Problem 13.7.5**

Find

$$\operatorname{Ln}(-11)$$

Solution

Problem 13.7.7

Find

$$\operatorname{Ln}(4 - 4i)$$

Solution

Problem 13.7.8

Find

$$\operatorname{Ln}(1 \pm i)$$

Solution

Problem 13.7.15

Find all values and graph some in the complex plane

$$\ln(e^i)$$

Solution

Problem 13.7.19

Solve for z

$$\ln z = 4 - 3i$$

Solution

Problem 13.7.23

Find the principal value.

$$(1 + i)^{1-i}$$

Solution

Problem 14.1.1

Find and sketch the path

$$z(t) = \left(1 + \frac{1}{2}i\right)t \quad (2 \leq t \leq 5)$$

Solution

Problem 14.1.11

Find a parametric representation and sketch the path

Segment from $(-1, 1)$ to $(1, 3)$

Solution

Problem 14.1.19

Find a parametric representation and sketch the path

Parabola $y = 1 - \frac{1}{4}x^2$ $(-2 \leq x \leq 2)$

Solution

Problem 14.1.21

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C \operatorname{Re} z \, dz, \quad C \text{ the shortest path from } 1 + i \text{ to } 3 + 3i$$

Solution

Problem 14.1.25

$$\int_C z \exp(z^2) \, dz, \quad C \text{ from } 1 \text{ along the axes to } i$$

Solution

Problem 14.1.29

$$\int_C \operatorname{Im} z^2 \, dz \quad \text{counterclockwise around the triangle with vertices } 0, 1, i$$

Solution